PyGameMaker Junior: Teacher Guide

What Junior Is

Junior is a simplified edition of PyGameMaker for students aged about 8 to 9. It is the same program and the same project files as the other editions. It only hides things, so a project made in Junior opens normally in Beginner, Advanced or Development, and the other way round.

Switching It On

1. Open **Tools > Preferences > General > IDE Edition** and choose **Junior**.
2. Click OK, then **restart the IDE**. The look, font and menus apply at startup.
3. To go back, repeat the steps and choose another edition. Preferences stays in the Tools menu in Junior for this reason.

Notes:

- **Shared computers:** the edition is a per-user setting. If you set up many machines, set "edition": "junior" in .pygamemaker/config.json in the user's home folder.
- **Theme and font:** if someone ever changed the theme or font in Preferences on a machine, that choice wins over Junior's larger font and high-contrast theme. Reset them there if the machine looks unchanged.
- **Blockly toolbox:** Junior's block set applies to **new** projects. An existing project keeps the block set it was saved with. Switching edition in Preferences also applies the new edition's block set to the open project.

What Students See

Area	In Junior
Look	Larger font, larger toolbar icons, high-contrast theme
File menu	New, Open, Save, Save As, Close, Recent, **Export as HTML5**, Exit
Assets menu	Import sprite, sound, background; **Create Sprite**, Object, Room
Build menu	**Test Game**
Tools menu	Preferences, Restore Deleted Assets, Language
Object editor	**Blockly only.** No Event List tab, no Code Editor tab
Blockly toolbox	46 blocks, exactly those the Junior tutorials use
Blockly right-click	Duplicate and Delete on blocks; Undo, Redo and Clean up on the workspace
Export	HTML5 only

Hidden on purpose: other exports (desktop, Android, Kivy, zip), GameMaker import, auto-save settings, Project Settings, Find and Replace, scripts and fonts, object and room packages, Debug and Build, Configure Action Blocks and the maintenance tools. Still visible: the small "Standard" events panel on the left of the object editor. It shows the events Blockly builds. Students can ignore it.

Suggested Lesson Plan (about 60 minutes)

Time	Activity
0-10	Tour: toolbar, resources, play button. Open a sample from the Welcome tab and press F5
10-25	Student handout, Parts 1 and 2: draw a sprite
25-45	Parts 3 and 4: make an object move, build a room, play
45-55	Part 5: export as HTML5 and open it in the browser
55-60	Share, and the "Think About It" questions as an exit ticket

For a 45-minute slot, do Parts 1 to 4 and leave the export for next time. Next lessons: the Junior tutorials under **Help > Tutorials**: Getting Started, Your First Game, Pong, Breakout, Fruit Fusion, Fruit Basket. Each has its own student handout and teacher guide.

Common Problems

Symptom	Cause and fix
Object does not move	No block under the event, or the object is not placed in the room
Hero slides forever	Needs **Keyboard: No key** with **Stop moving**
Sprite not visible	Object has no sprite chosen
"Name Already Used"	A sprite with that name exists. Choose another name
A name is rejected	Use letters, numbers and underscores, with no spaces or special characters
Deleted by mistake	Tools > Restore Deleted Assets
Grey game window	Often a slow or low-memory computer. Wait, then retry
Exported game needs internet	Games using Python code load a runtime online. When asked, choose to bundle it for offline use
A sample uses a block students cannot find	Samples may use blocks outside Junior's set. They still run. Students just cannot add that block again

Limits to Know

- Students cannot add the blocks that Beginner has and Junior does not, for example Set Gravity, If Collision and Sleep, even when a sample uses them.
- A sprite starts at 32 by 32 pixels. Students can resize it in the sprite editor.
- HTML5 is the only way to share. A teacher who needs another format can switch the edition to Beginner and use **Build > Export Game**.

Checking Work

Ask each student to show: one sprite they drew, one object that moves, a room, and the exported .html file running in the browser.